Lecture 2

Real Business Cycle Models

Version 1.2
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Changes from version 1.0 are in red

Changes from version 1.0 are in purple

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1 Introduction

- We now know more about business cycle facts (my first lecture).
- Let us consider a simple stochastic version of an optimal growth model .
- Let's see how "simple" versions of this model compare with the business cycle facts.
- Note that the goal here is to produce *quantitative* predictions.

2 A First Analytical Model

- This model is a version of Brock and Mirman (1972, JET)
- Brock and Mirman provided the first optimizing growth model with stochastic shocks.

2.1 Setup

- Social planner solves

$$\max \quad \mathrm{E}_t \sum_{n=0}^{\infty} \beta^n \log C(s^{t+n})$$

subject to

$$\begin{aligned} Y_t(s^t) &= Z_t(s^t)\left(K_t(s^{t-1})\right)^\alpha \\ K_{1+1}(s^t) &= Y_t(s^t) - C_t(s^t) \end{aligned}$$

with $0 < \alpha < 1$, Z_t follows some stochastic markovian process and $s_t = \{Z_t, K_t\}$.

- Note 3 important assumptions:

  1. No labor supply decision
  2. log utility
  3. full depreciation ($K_{t+1} = (1-\delta)K_t + I_t$ with $\delta = 1$).

- Let's rewrite the problem in terms of Bellman's equation:

$$V(K_t, Z_t) = \max \quad \log C_t + \beta E_t V(K_{t+1}, Z_{t+1})$$

subject to

$$K_{t+1} = Z_t K_t^{\alpha} - C_t$$

2.2 First Order Conditions

- Replace in the Bellman's equation K_{t+1} by its value in the resource constraint and derive to obtain:

$$\frac{1}{C_t} = \beta E_t \frac{\partial V\big(K_{t+1}, Z_{t+1}\big)}{\partial K_{t+1}} \tag{1}$$

$$\frac{\partial V\big(K_t, Z_t\big)}{\partial K_t} = \frac{1}{C_t}\alpha Z_t K_t^{\alpha-1} \tag{2}$$

- Substituting:

$$\frac{1}{C_t} = \beta E_t \left[\frac{1}{C_{t+1}}\alpha Z_{t+1} K_{t+1}^{\alpha-1}\right]$$

$$\Longleftrightarrow$$

$$1 = \beta E_t \left[\frac{C_t}{C_{t+1}} \alpha Z_{t+1} K_{t+1}^{\alpha-1} \right] \tag{3}$$

- Note that this look like the standard consumption Euler equation of the permanent income model

$$1 = \beta E_t \left[R_{t+1} \frac{C_t}{C_{t+1}} \right]$$

with $R_{t+1} = \alpha Z_{t+1} K_{t+1}^{\alpha-1}$.

2.3 Solution

- Let us guess a solution of the form $C_t = \gamma Y_t$ (constant saving rate)
- Let's verify that the Euler equation (3) is then satisfied (for a specific value of γ that we need to find).
- (3) writes

$$1 = \beta E_t \left[\frac{\gamma Y_t}{\gamma Y_{t+1}} \alpha Z_{t+1} K_{t+1}^{\alpha-1} \right]$$

- Note that $Z_{t+1} K_{t+1}^{\alpha-1} = \frac{Y_{t+1}}{K_{t+1}}$ so that

$$1 = \beta E_t \left[\frac{Y_t}{Y_{t+1}} \alpha \frac{Y_{t+1}}{K_{t+1}} \right]$$

$$\Longleftrightarrow$$

$$\begin{aligned} 1 &= \beta E_t \left[\alpha \frac{Y_t}{K_{t+1}} \right] \\ &= \beta E_t \left[\alpha \frac{Y_t}{Y_t - C_t} \right] \\ &= \beta E_t \left[\alpha \frac{Y_t}{(1-\gamma) Y_t} \right] \\ &= \frac{\alpha\beta}{1-\gamma} \end{aligned}$$

which implies that $\gamma = 1 - \alpha\beta$

- Then $K_{t+1} = (1-\gamma)Y_t = \alpha\beta Z_t K_t^{\alpha}$.
- In logs : $k_{t+1} = \log \alpha\beta + z_t + \alpha k_t$.

$$k_{t+1} = \log \alpha\beta + z_t + \alpha k_t$$

- This last equation shows that there are two sources of dynamics in the model:

1. The dynamics of the exogenous shock z
2. The own dynamics of k, which is related to consumption smoothing: the social planner will never choose to consume 100% of a shock on impact, but will aim at smoothing consumption using savings (capital).

- Using the equation $K_{t+1} = \alpha\beta Y_t$, one gets the dynamics of output:

$$y_{t+1} = \alpha y_t + z_{t+1} + \alpha \log \alpha\beta$$

2.4 Impulse Response to a Permanent Shock

- Assume z follows a random walk

$$z_{t+1} = z_t + \varepsilon_{t+1}$$

- Assume the economy is before $t = 1$ at its non stochastic steady-state $\overline{y}$ (defined as the steady state when $\varepsilon_t = 0\ \forall t$).
- $\overline{y} = \frac{\alpha}{1-\alpha} \log \alpha\beta$.
- The process of y then writes

$$y_{t+1} = \alpha y_t + z_{t+1} + (1 - \alpha)\overline{y}$$

- Assume $\varepsilon_1 = 1$ and $\varepsilon_t = 0\ \forall t > 1$.

- Then

$$y_1 = \overline{y} + 1$$

$$y_2 = \overline{y} + 1 + \alpha$$

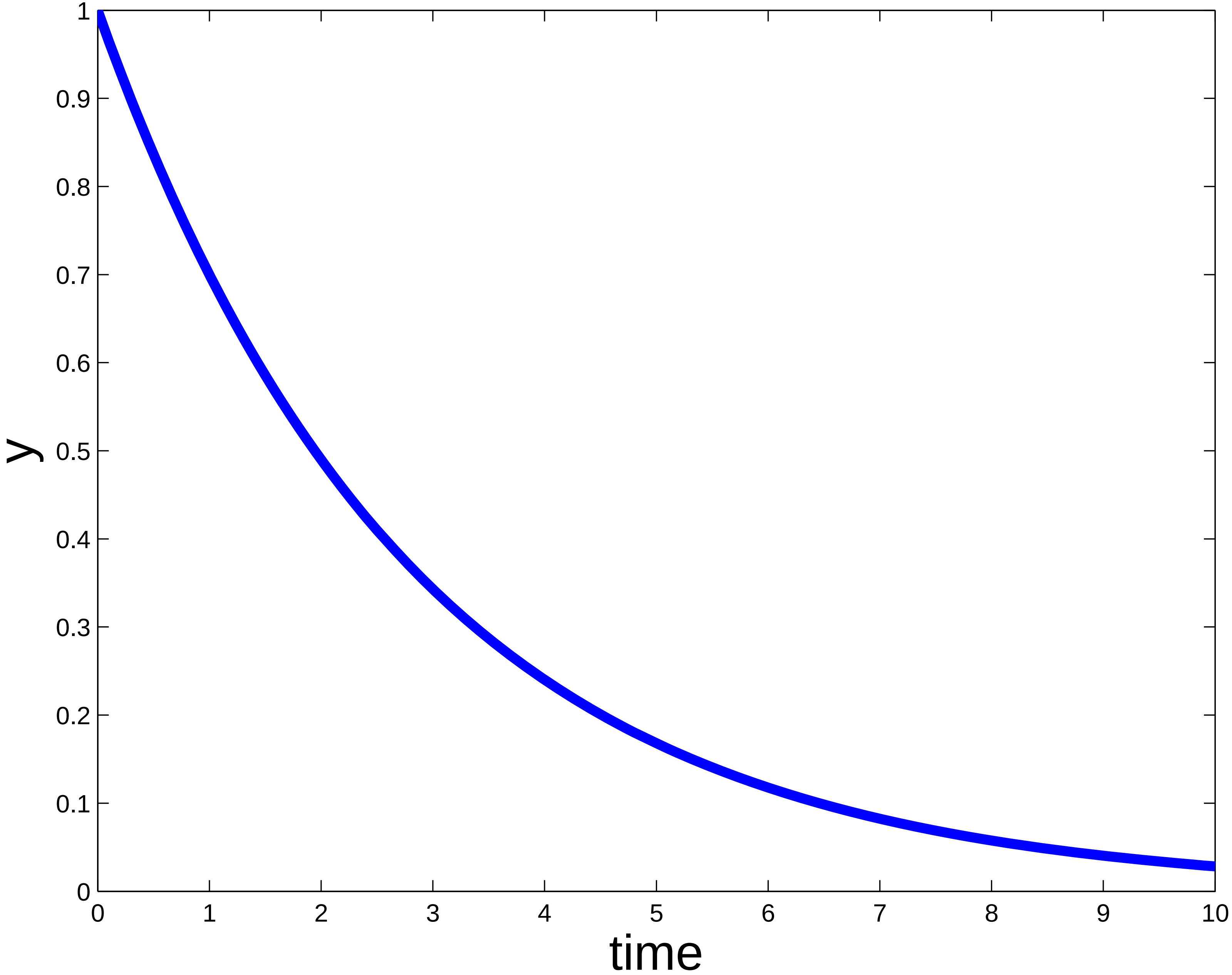
$$y_2 = \overline{y} + 1 + \alpha + \alpha^2$$

$$....$$

$$y_t = \overline{y} + 1 + \alpha + \alpha^2 + \ldots + \alpha^t$$

$$....$$

Figure 1: Response to a permanent shock



2.5 Impulse Response to a Transitory Shock

- Assume now that z is a white noise

$$z_t = \varepsilon_{t+1}$$

- Assume the economy is before $t = 1$ at its non stochastic steady-state $\overline{y}$ (defined as the steady state when $\varepsilon_t = 0\ \forall t$).
- Assume $\varepsilon_1 = 1$ and $\varepsilon_t = 0\ \forall t > 1$.
- Then

$$y_1 = \overline{y} + 1$$

$$y_2 = \overline{y} + \alpha$$

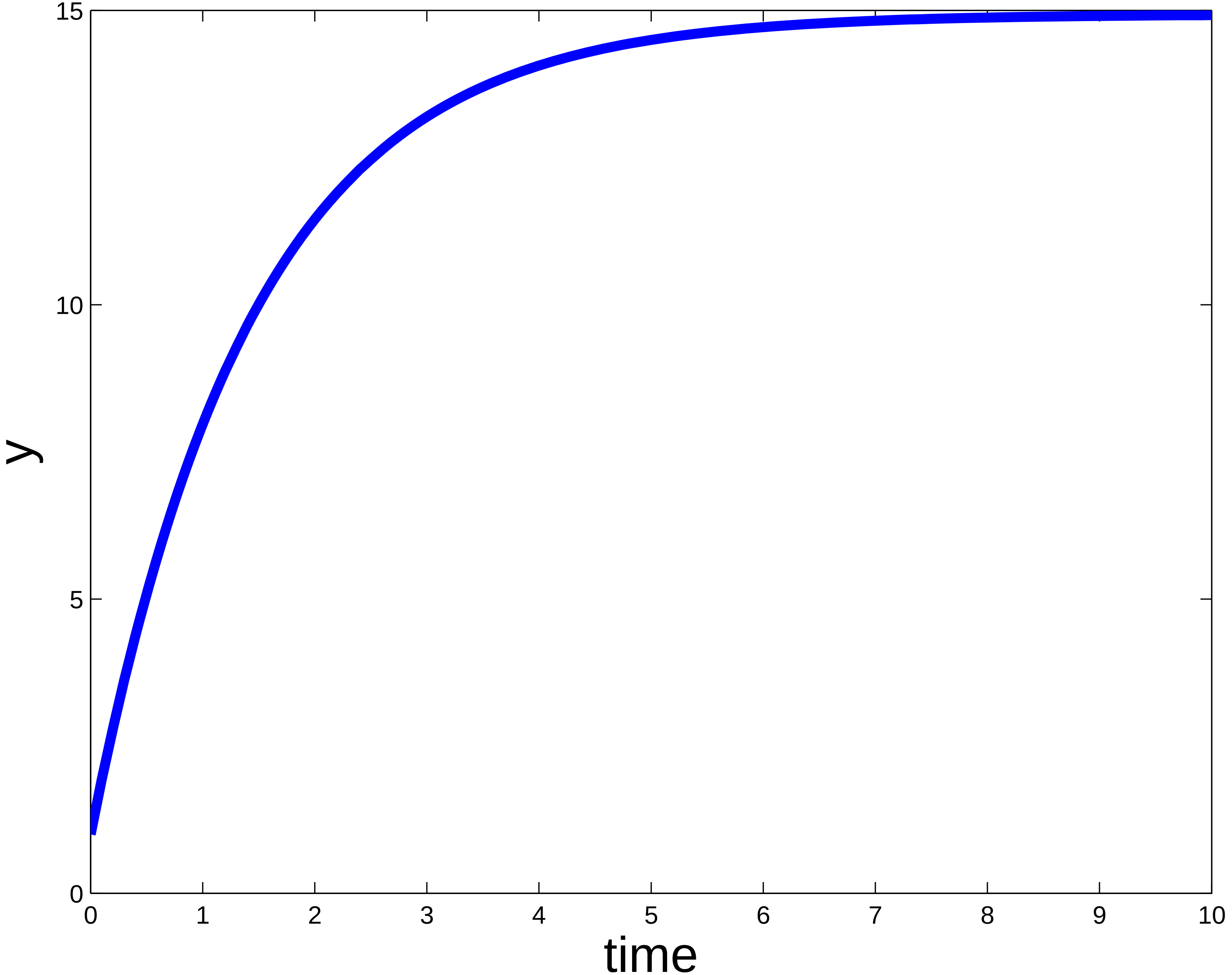
$$y_2 = \overline{y} + \alpha^2$$

....

$$y_t = \overline{y} + \alpha^t$$

....

Figure 2: Response to a temporary shock



3 The Standard Real Business Cycle (RBC) Model

- Perfectly competitive economy
- Optimal growth model + Labor decisions
- 2 types of agents: Households and Firms
- Shocks to productivity
- Pareto optimal economy
- We will solve for the competitive equilibrium

3.1 The Household

- Mass of agents = 1 (no population growth)
- Identical agents + All face the same aggregate shocks (no idiosyncratic uncertainty)
- $\leadsto$ One representative household

- Infinitely lived rational agent with intertemporal utility

$$E_t \sum_{s=0}^{\infty} \beta^s U_{t+s}$$

$\beta \in (0,1)$: discount factor,

- Preferences over
  - a consumption bundle
  - leisure

- $U_t = U(C_t, \ell_t)$ with $U(\cdot,\cdot)$

  × class $\mathcal{C}^2$, strictly increasing, concave and satisfy Inada conditions

  × compatible with balanced growth [more below]:

$$U(C_t, \ell_t) = \begin{cases} \frac{C_t^{1-\sigma}}{1-\sigma} v(\ell_t) & \text{for } \sigma \in \mathbb{R}^+ \backslash \{1\} \\ \text{or} & \\ \log(C_t) + v(\ell_t) & \end{cases}$$

Preferences are therefore given by

$$E_t \left[\sum_{s=0}^{\infty} \beta^s U(C_{t+s}, \ell_{t+s}) \right]$$

- Household faces three constraints

- Time constraint

$$h_{t+s} + \ell_{t+s} \leqslant T = 1$$

(for convenience T=1)

- Budget constraint

$$\underbrace{B_{t+s}}_{Bond\ purchases} + \underbrace{C_{t+s} + I_{t+s}}_{Good\ purchases}$$

$$\leqslant \underbrace{(1 + r_{t+s-1})B_{t+s-1}}_{Bond\ revenus} + \underbrace{W_{t+s}h_{t+s}}_{Wages} + \underbrace{z_{t+s}K_{t+s}}_{Capital\ revenus}$$

- Capital Accumulation

$$K_{t+s+1} = I_{t+s} + (1 - \delta)K_{t+s}$$

$\delta \in (0, 1)$: depreciation rate

The household decides on consumption, labor, leisure, investment, bond holdings and capital formation maximizing utility constraint, taking the constraints into account

$$\max_{\{C_{t+s},h_{t+s},\ell_{t+s},I_{t+s},K_{t+s+1},B_{t+s}\}_{s=0}^{\infty}} E_t \left[\sum_{s=0}^{\infty} \beta^s U\big(C_{t+s}, \ell_{t+s}\big) \right]$$

subject to the sequence of constraints

$$\begin{cases} h_{t+s} + \ell_{t+s} \leqslant 1 \\ B_{t+s} + C_{t+s} + I_{t+s} \leqslant \big(1 + r_{t+s-1}\big) B_{t+s-1} + W_{t+s} h_{t+s} + z_{t+s} K_{t+s} \\ K_{t+s+1} = I_{t+s} + (1-\delta) K_{t+s} \\ K_t,\ B_{t-1} \text{ given} \end{cases}$$

$$\max_{\{C_{t+s},h_{t+s},K_{t+s+1},B_{t+s}\}_{s=0}^{\infty}} E_t\left[\sum_{s=0}^{\infty}\beta^s U(C_{t+s},1-h_{t+s})\right]$$

subject to

$$B_{t+s}+C_{t+s}+K_{t+s+1}\leqslant(1+r_{t+s-1})B_{t+s-1}+W_{t+s}h_{t+s}+(z_{t+s}+1-\delta)K_{t+s}$$

Write the Lagrangian

$$\mathcal{L}_t = E_t\sum_{s=0}^{\infty}\beta^s\Bigg[U(C_{t+s},1-h_{t+s})+\Lambda_{t+s}\Bigg((1+r_{t+s-1})B_{t+s-1}+$$

$$+W_{t+s}h_{t+s}+(z_{t+s}+1-\delta)K_{t+s}-C_{t+s}-B_{t+s}-K_{t+s+1}\Bigg)\Bigg]$$

First order conditions ($\forall\ s \geq 0$)

$$\begin{aligned}
C_{t+s} &: E_t U_c\big(C_{t+s}, 1 - h_{t+s}\big) = E_t \Lambda_{t+s} \\
h_{t+s} &: E_t U_\ell\big(C_{t+s}, 1 - h_{t+s}\big) = E_t\big(\Lambda_{t+s} W_{t+s}\big) \\
B_{t+s} &: E_t \Lambda_{t+s} = \beta E_t\big((1 + r_{t+s})\Lambda_{t+s+1}\big) \\
K_{t+s+1} &: E_t \Lambda_{t+s} = \beta E_t\big(\Lambda_{t+s+1}\big(z_{t+s+1} + 1 - \delta\big)\big)
\end{aligned}$$

and the transversality condition

$$\lim_{s \longrightarrow +\infty} \beta^s E_t \Lambda_{t+s}\big(B_{t+s} + K_{t+s+1}\big) = 0$$

Remark :

- It is convenient to write and interpret FOC for $s = 0$:

$$\begin{array}{ll} h_t & : U_\ell(C_t, 1-h_t) = U_c(C_t, 1-h_t)W_t \\ B_t & : U_c(C_t, 1-h_t) = \beta\big((1+r_t)E_t U_c(C_{t+1}, 1-h_{t+1})\big) \\ K_{t+1} & : U_c(C_t, 1-h_t) = \beta E_t\big(U_c(C_{t+1}, 1-h_{t+1})(z_{t+1}+1-\delta)\big) \end{array}$$

and the transversality condition

$$\lim_{s \longrightarrow +\infty} E_t \beta^s U_c(C_{t+s}, 1-h_{t+s})\left(K_{t+s+1} + B_{t+s}\right) = 0$$

Simple example : Assume $U(C_t, \ell_t) = \log(C_t) + \theta \log(1 - h_t)$

$$
\begin{array}{ll}
h_{t+s} & : E_t \frac{1}{1-h_{t+s}} = E_t \frac{W_{t+s}}{C_{t+s}} \\
B_{t+s} & : E_t \frac{1}{C_{t+s}} = \beta E_t (1 + r_{t+s}) \frac{1}{C_{t+s+1}} \\
K_{t+s+1} & : E_t \frac{1}{C_{t+s}} = \beta E_t \frac{1}{C_{t+s+1}} (z_{t+s+1} + 1 - \delta)
\end{array}
$$

and the transversality condition

$$
\lim_{s \longrightarrow +\infty} E_t \beta^s \frac{K_{t+s+1} + B_{t+s}}{C_{t+s}} = 0
$$

3.2 The Firm

- Mass of firms = 1
- Identical firms + All face the same aggregate shocks (no idiosyncratic uncertainty)

$\rightsquigarrow$ Representative firm

- Produce an homogenous good that is consumed or invested
- by means of capital and labor

- Constant returns to scale technology (important)

$$Y_t = A_t F(K_t, \Gamma_t h_t)$$

- $\Gamma_t = \gamma \Gamma_{t-1}$ Harrod neutral technological progress ($\gamma \geqslant 1$), A_t stationary (does not explain growth)

- Remark: one could introduce long run technical progress in three different ways:

$$Y_t = \widehat{\Gamma}_t F(\widetilde{\Gamma}_t K_t, \Gamma_t h_t)$$

- $\widehat{\Gamma}_t$ is Hicks Neutral, $\widetilde{\Gamma}_t$ is Solow neutral, Γ_t is Harrod neutral

- Harrod neutral technical progress and the preferences specified above are needed for the existence of a *Balanced Growth Path* that replicates *Kaldor Stylized Facts*:

  1. The shares of national income received by labor and capital are roughly constant over long periods of time
  2. The rate of growth of the capital stock is roughly constant over long periods of time
  3. The rate of growth of output per worker is roughly constant over long periods of time
  4. The capital/output ratio is roughly constant over long periods of time
  5. The rate of return on investment is roughly constant over long periods of time
  6. The real wage grows over time

- For preferences, what is needed is wealth and substitution effect to cancel out in labor supply

$$Y_t = A_t F(K_t, \Gamma_t h_t)$$

- $\Gamma_t = \gamma \Gamma_{t-1}$ Harrod neutral technological progress ($\gamma \geqslant 1$), A_t stationary (does not explain growth)

- A_t are shocks to technology. AR(1) exogenous process

$$\log(A_t) = \rho \log(A_{t-1}) + (1-\rho)\log(\overline{A}) + \varepsilon_t$$

with $\varepsilon_t \leadsto \mathcal{N}(0, \sigma^2)$.

The firm decides on production plan maximizing profits

$$\max_{\{K_t,h_t\}} A_t F(K_t, \Gamma_t h_t) - W_t h_t - z_t K_t$$

First order conditions:

$$\begin{aligned} K_t &: A_t F_K(K_t, \Gamma_t h_t) = z_t \\ h_t &: A_t F_h(K_t, \Gamma_t h_t) = W_t \end{aligned}$$

Simple Example: Cobb–Douglas production function

$$Y_t = A_t K_t^{\alpha} (\Gamma_t h_t)^{1-\alpha}$$

First order conditions

$$\begin{aligned} K_t &: \alpha Y_t / K_t = z_t \\ h_t &: (1-\alpha) Y_t / h_t = W_t \end{aligned}$$

3.3 Equilibrium

The (RBC) Model Equilibrium is given by the following equations ($\forall\ t \geq 0$):

1. Exogenous Processes : $\log(A_t) = \rho \log(A_{t-1}) + (1-\rho)\log(\overline{A}) + \varepsilon_t$ and $\Gamma_t = \gamma \Gamma_{t-1}$
2. Law of motion of Capital : $K_{t+1} = I_t + (1-\delta)K_t$
3. Bond market equilibrium : $B_t = 0$
4. Good Markets equilibrium : $Y_t = C_t + I_t$

5. Labor market equilibrium : $\frac{U_\ell(C_t,1-h_t)}{U_c(C_t,\ell_t)} = A_t F_h(K_t, \Gamma_t h_t)$

6. Consumption/saving decision + Capital market equilibrium :

$$U_c(C_t, 1-h_t) = \beta E_t \left[U_c(C_{t+1}, 1-h_{t+1})(A_{t+1}F_K(K_{t+1}, \Gamma_{t+1}h_{t+1}) + 1 - \delta)\right]$$

7. Financial markets :

$$(1+r_t) = \frac{U_c(C_t, 1-h_t)}{\beta E_t U_c(C_{t+1}, 1-h_{t+1})}$$

3.4 An Analytical Example

- $U(C_t, \ell_t) = \log(C_t) + \theta \log(\ell_t), \qquad Y_t = A_t K_t^{\alpha} (\Gamma_t h_t)^{1-\alpha}$

- Equilibrium

$$\frac{\theta C_t}{1 - h_t} = (1-\alpha)\frac{Y_t}{h_t}$$
$$\frac{1}{C_t} = \beta E_t \left[\frac{1}{C_{t+1}} \left(\alpha \frac{Y_{t+1}}{K_{t+1}} + 1 - \delta \right) \right]$$
$$K_{t+1} = Y_t - C_t + (1-\delta) K_t$$

$$Y_t = A_t K_t^{\alpha} (\Gamma_t h_t)^{1-\alpha}$$

$$\lim_{s \longrightarrow \infty} \beta^s E_t \left[\frac{K_{t+1+s}}{C_{t+s}} \right] = 0$$

3.5 Stationarization

- We want a stationary equilibrium (technical reasons)
- Deflate the model for the growth component Γ_t
- On the example: $x_t = X_t/\Gamma_t$

- Deflated Equilibrium

$$\frac{\theta c_t}{1-h_t} = (1-\alpha)\frac{y_t}{h_t}$$
$$\frac{1}{c_t} = \frac{\beta}{\gamma}E_t\left[\frac{1}{c_{t+1}}\left(\alpha\frac{y_{t+1}}{k_{t+1}} + 1 - \delta\right)\right]$$
$$\gamma k_{t+1} = y_t - c_t + (1-\delta)k_t$$
$$y_t = A_t k_t^{\alpha} h_t^{1-\alpha}$$
$$\lim_{s\longrightarrow\infty} \beta^s \frac{\gamma k_{t+1+s}}{c_{t+s}} = 0$$

3.6 Solving the Model

3.6.1 In General

- Non–linear system of stochastic finite difference equations under rational expectations
- Very complicated
- In general no analytical solution, need to rely on numerical approximation methods

3.6.2 The Nice Analytical Case

- $U(c, \ell) = \log(c) + \theta \log(\ell)$, $\gamma = 1$ (not needed) and $\delta = 1$
- This is the case we solved in section 2 of this lecture.

3.6.3 Numerical solution

- Unfortunately: No closed form solution in general
- Have to adopt a numerical approach
- Log–linearize the equilibrium around the steady state (limits?)

- Solve the linearized model using standard techniques

- The solution to the log–linearize version of the model takes the form

$$X_{t+1} = M_x X_t + M_z Z_t \tag{4}$$

$$Z_{t+1} = \rho Z_t + \varepsilon_{t+1} \tag{5}$$

$$Y_t = P_x X_t + P_z Z_t \tag{6}$$

where X_t, Z_t and Y_t collect, respectively, the state variables, the shocks and the variables of interest

- Resembles a VAR model

- In the basic RBC model

$$X_t = \{k_t\} \tag{7}$$

$$Z_t = \{a_t\} \tag{8}$$

$$Y_t = \{y_t, c_t, i_t, h_t\} \tag{9}$$

- Let's evaluate the **quantitative** ability of the model to account for and explain the cycle.

3.7 Quantitative Evaluation

3.7.1 Calibration

- Need to assign numerical values to the parameters
- This is the **calibration** step
- What does calibration mean?
  - × Make explicit use of the model to set the parameters
  - × A lot of discipline, but no systematic recipe
  - × Have to set $(\alpha, \beta, \theta, \delta, \gamma, \rho, \sigma, \overline{A})$
  - × Use data: $(\Delta \log Y, k/y, i/k, h, wh/y)$

- Compute $(\Delta \log Y, k/y, i/k, h, wh/y)$ in the model to set $(\alpha, \beta, \theta, \delta, \gamma)$
- Then compute A_t on the data and estimate ρ and σ.

- In the data $\Delta y = 0.9\%$ per quarter $\leadsto \gamma = 1.009$.

- In the data $i/k = 0.076$ on annual data. Use capital accumulation to get annual depreciation.

$$i/k(\text{model}) = \gamma - (1-\delta) \leadsto \delta = 0.01$$

- In the data $wh/y = 0.6$, in the model $wh/y = 1-\alpha \leadsto \alpha = 0.4$.

- In the data $k/y = 3.32$ on annual data. Using the Euler equation

$$\beta = \frac{\gamma}{\alpha y/k + (1-\delta)}$$

which gives $\beta = 0.98$

- $h = 0.31$, such that $\theta = \frac{(1-\alpha)(1-h)}{hc/y}$

- We set $\overline{A} = 1$ without loss of generality.

- We have $\log(A_t) = \log(y_t) - \alpha \log(k_t) - (1-\alpha)\log(h_t)$

- Estimate $\log(A_t) = \rho \log(A_{t-1}) + \varepsilon_t$

- We obtain $\rho = 0.95$ and $\sigma = 0.0079$.

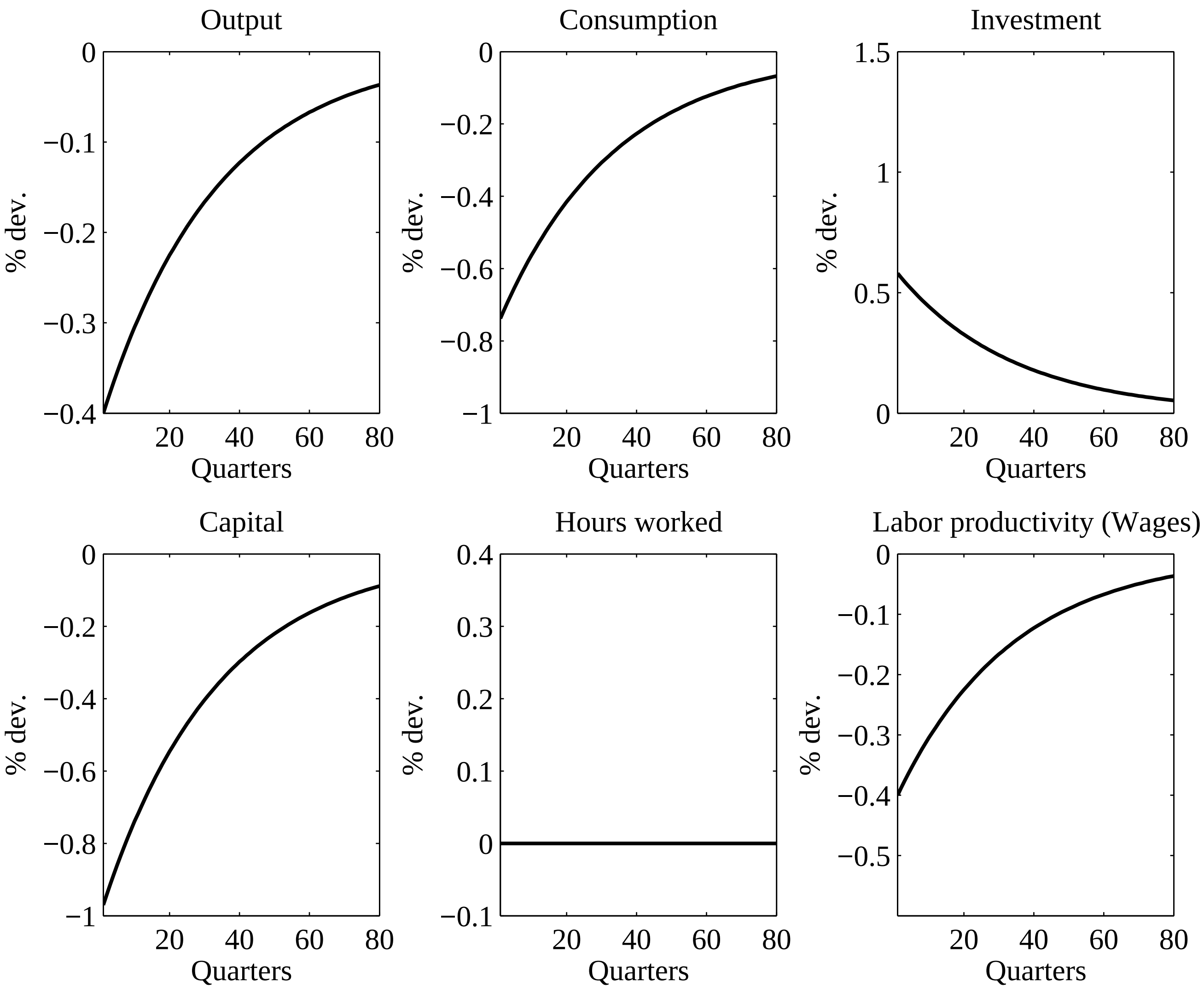
3.7.2 Can the Business Cycle be Driven by Capital Dynamics?

- Want to see whether capital dynamics can account for BC.

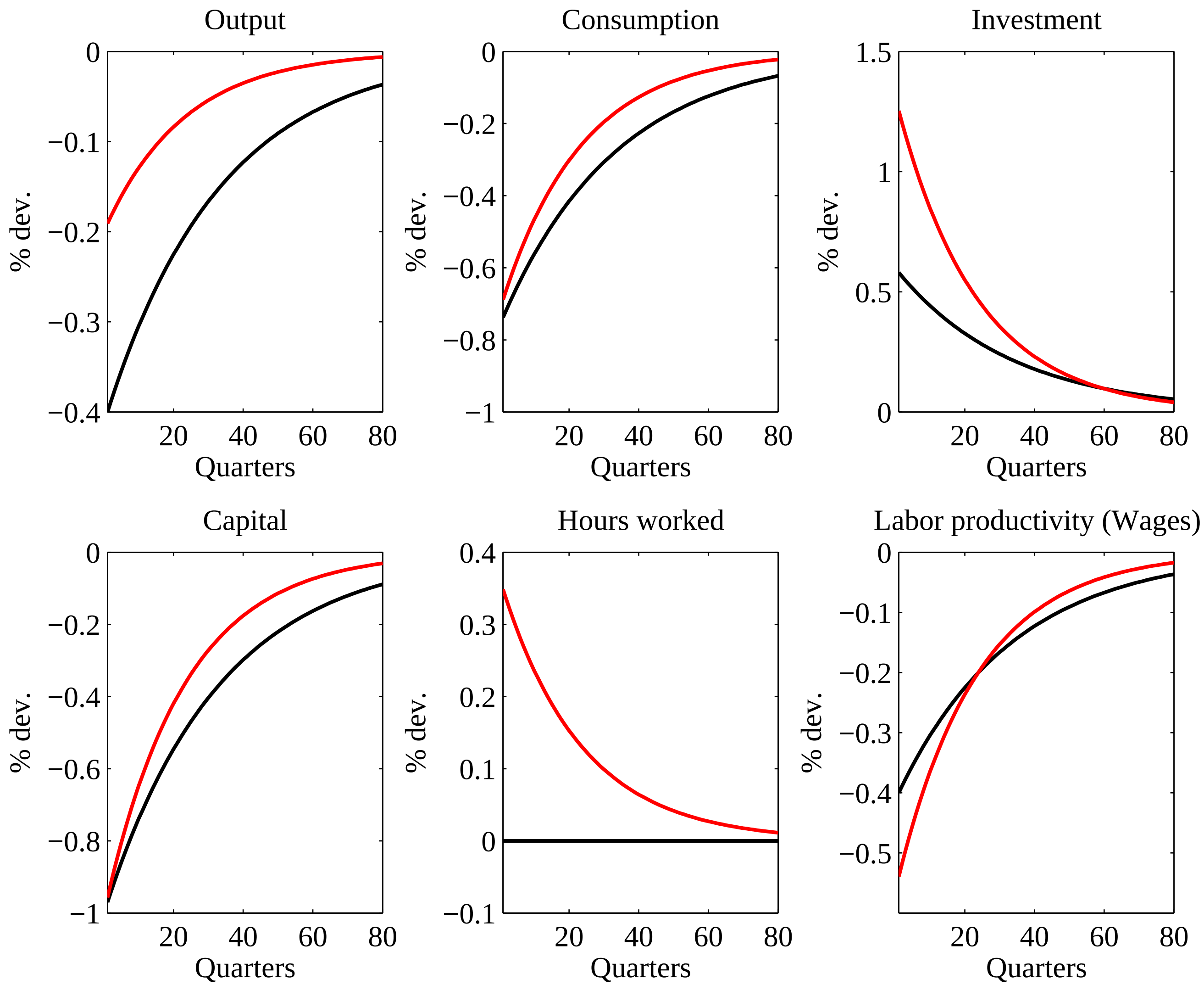
- Perfect foresights dynamics

- We study a case with fixed labor $(h = \overline{h})$ and a variable labor case $(U(c, 1-h))$

Capital shock - Fixed hours



Capital shock - Variable hours

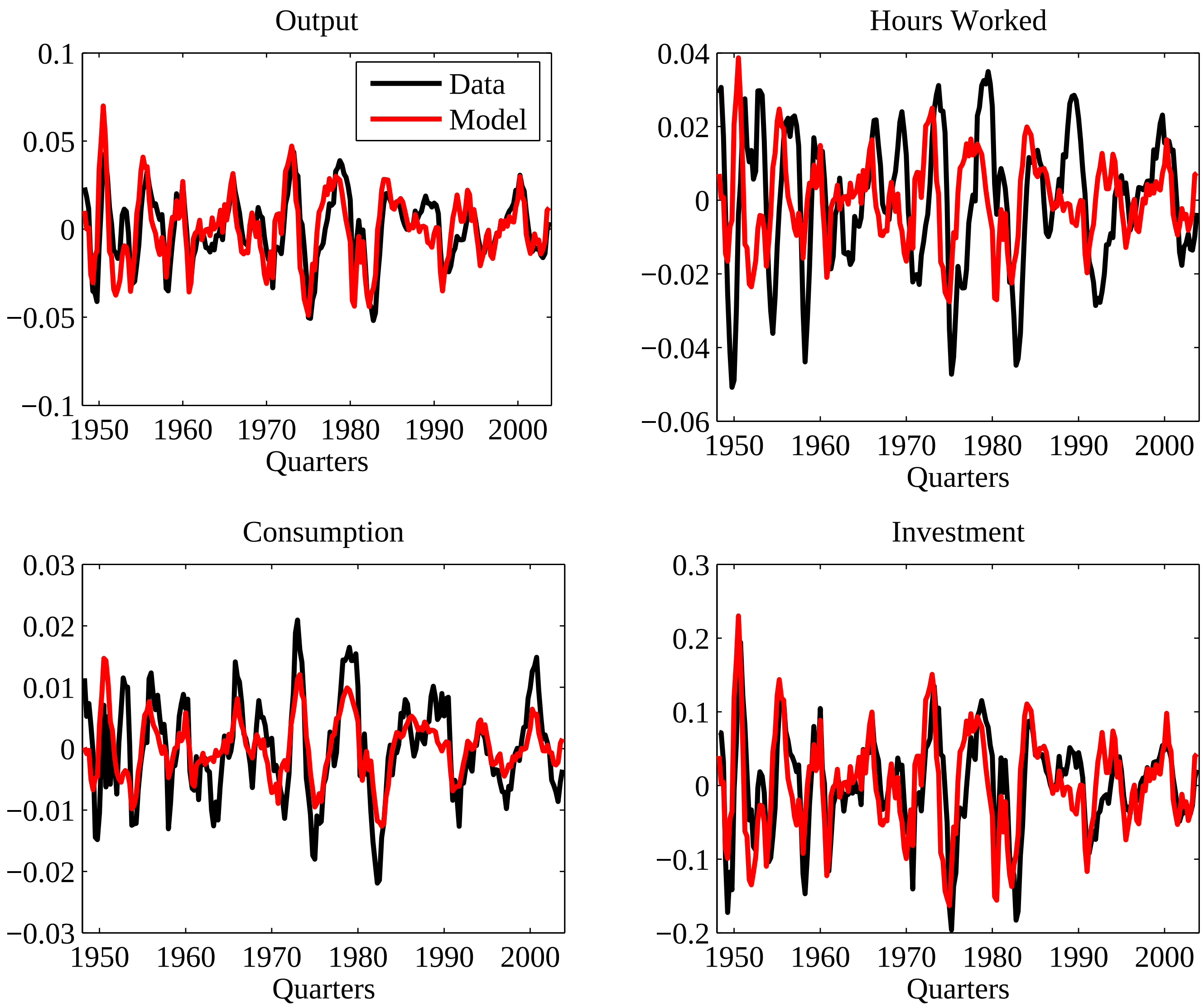


- Capital dynamics cannot be the story for the business cycle.
- More (shocks?) is needed to understand the BC
- That's why the RBC literature proposes technological shocks (Brock & Mirman, 1972)

3.7.3 Technological Shocks

- We first stick in the model the estimated ε_t and compare model and data.

A good fit with estimated shocks



A first success?

- Accounts for the main events in the data
- The model correctly predicts the data: corr(y, y^m)=0.75, corr(c, c^m)=0.73, corr(i, i^m)=0.70.
- BUT: corr(h, h^m)=0.06
- Let's compute unconditional moments in the model (simulate-HP filter-compute moments)

3.7.4 Results from Model Simulations

- Let's first look at data characteristics.

U.S. data, 1948-2005

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	Auto(1)
Output	1.70	–	–	–	0.84
Consumption	0.80	0.47	0.78	–	0.83
Services	1.11	0.66	0.72	–	0.80
Non Durables	0.72	0.42	0.71	–	0.77
Investment	6.49	3.83	0.84	–	0.81
Fixed investment	5.08	3.00	0.80	–	0.88
Durables	5.23	3.09	0.58	–	0.72
Changes in inventories	22.48	13.26	0.48	–	0.40
Hours worked	1.69	1.00	0.86	–	0.89
Labor productivity	0.90	0.53	0.41	0.09	0.69

- Now consider the results from model simulations

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	Auto
Output	1.70	–	–	–	0.84
	1.49	–	–	–	0.68
Consumption	0.80	0.47	0.78	–	0.83
	0.37	0.25	0.81	–	0.82
Investment	6.49	3.83	0.84	–	0.81
	5.00	3.35	0.99	–	0.68
Hours worked	1.69	1.00	0.86	–	0.89
	0.85	0.57	0.98	–	0.68
Labor productivity	0.90	0.53	0.41	0.09	0.69
	0.67	0.45	0.97	0.92	0.72

(model in yellow)

3.7.5 A success(?)

- The model correctly predicts the amplitude, serial correlation and relative variability of fluctuations
- It accounts for a large part of output volatility
- Correct ranking of the volatility of c, i, y, ...
- Large serial correlation, although it is smaller than in the data.
- **But**

- × C and N are not volatile enough
- × w (and r) are to too procyclical

3.8 Criticisms

3.8.1 In General

- The research on RBC became so successful because
  - Propose a coherent platform to analyse growth and cycles
  - It somewhat fails such that there is room for work
- Main victory: methodological (part of the toolkit of macroeconomists)
- Main criticisms
  - incorrect/implausible calibration of parameters (IES, Labor

supply) $\rightsquigarrow$ need for sensibility analysis

- counterfactual prediction for some prices:
  - × real wage is strongly procyclical in the model,
  - × CRRA preferences are not compatible with the equity premium
  - × price level is too strongly countercyclical

3.8.2 The Measure of Technological Shocks

- Key problem: Are technological shocks at the source of BC?

- Prescott [1986]: Technology shocks account for 70% of output volatility, but
  - × Too volatile
  - × Little evidence of large supply shocks (except oil prices)
  - × Recessions have to be explained by technological regressions
  - × Measurement problems (contamination by demand shocks if increasing returns, imperfect competition, labor hoarding) ; in the growth accounting literature, the SR was

a measure of our ignorance, now it is the engine of the model.

3.8.3 The Need for Persistent Shocks

- Recall that

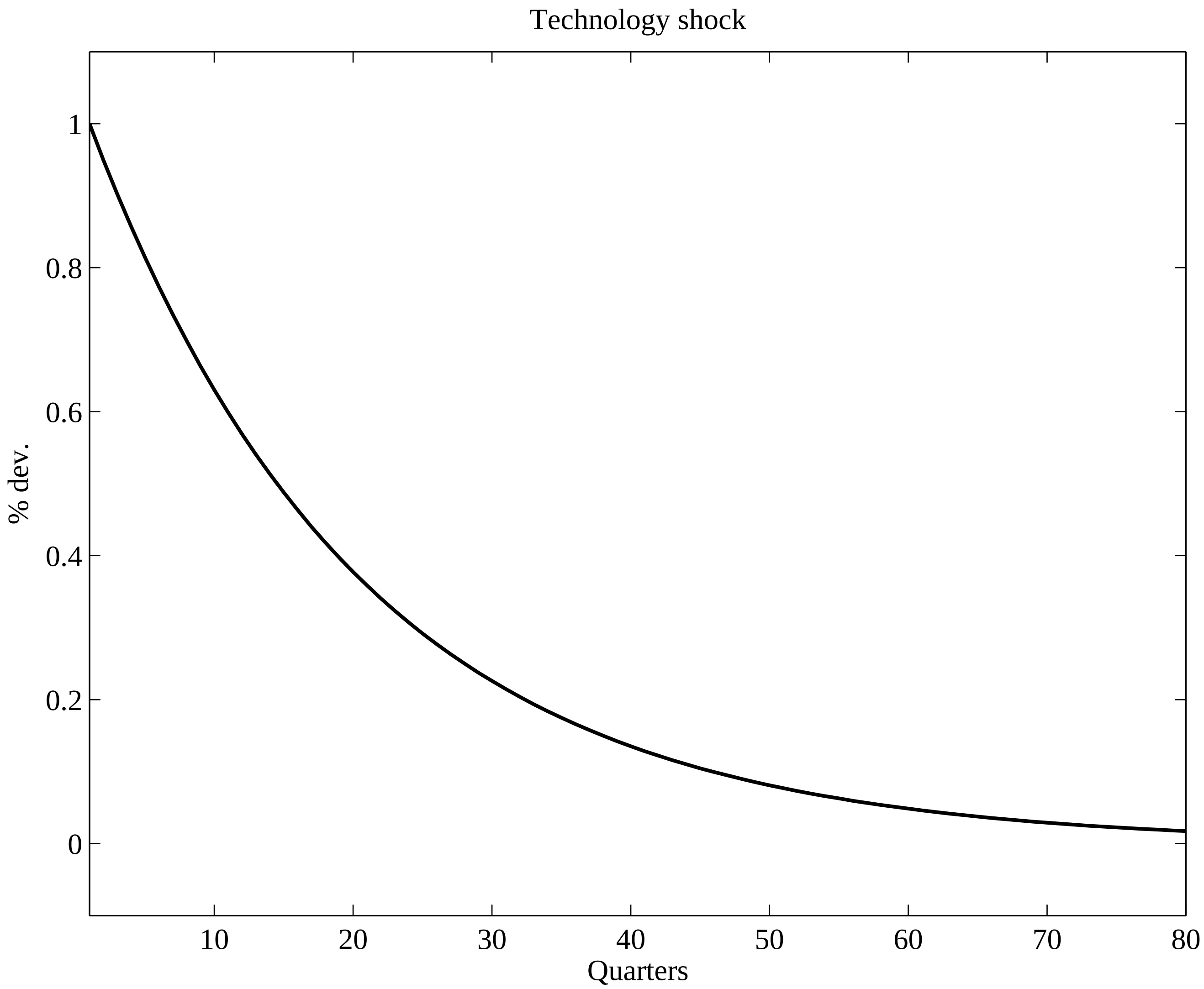
$$\log(A_t) = \rho \log(A_{t-1}) + \varepsilon_t$$

  and ρ is large

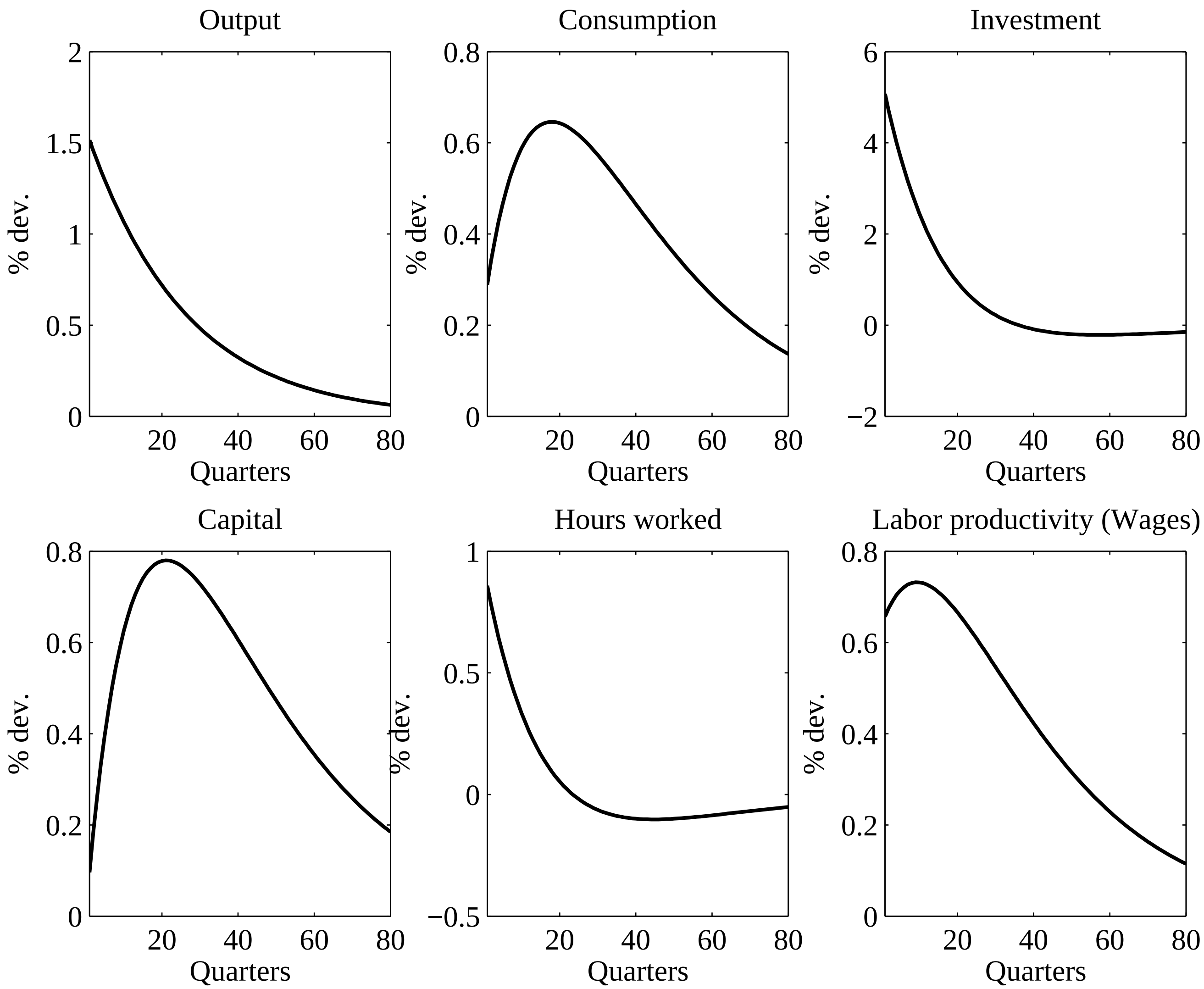
- Why do we need so persistent shocks?
- Because the model possesses weak propagation mechanisms

- Let's see that in details

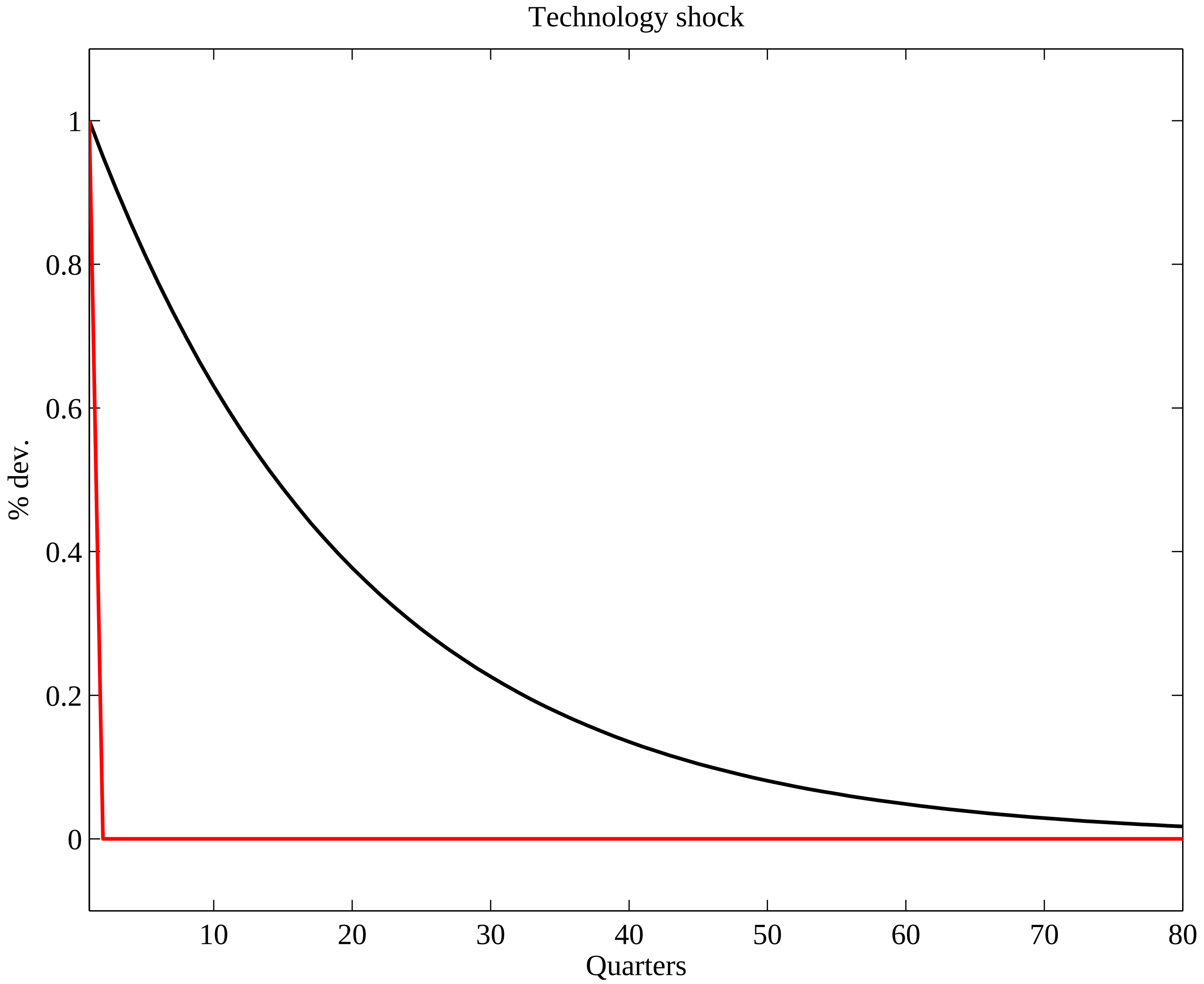
IRF to A Technological Shock



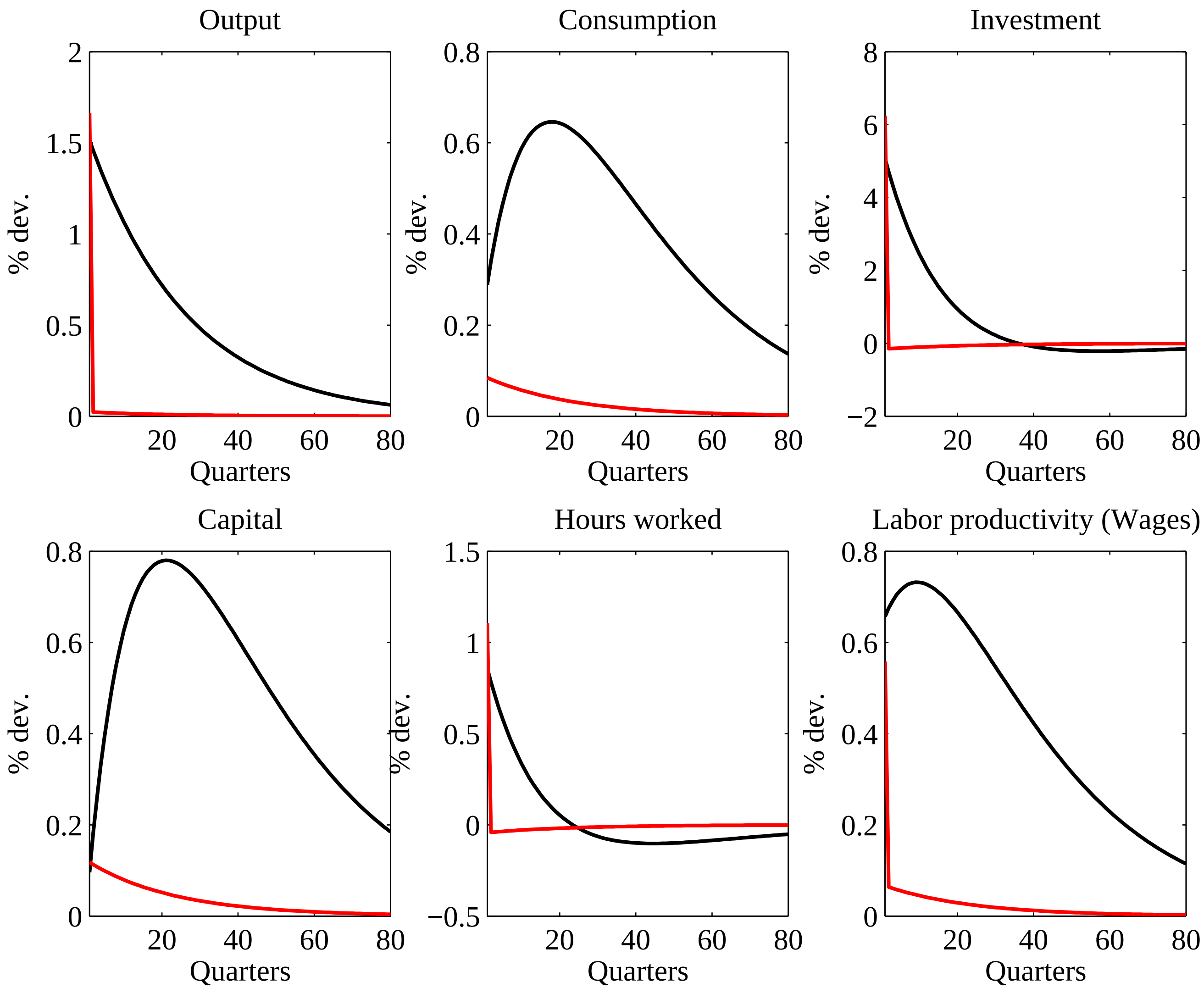
IRF to A Technological Shock



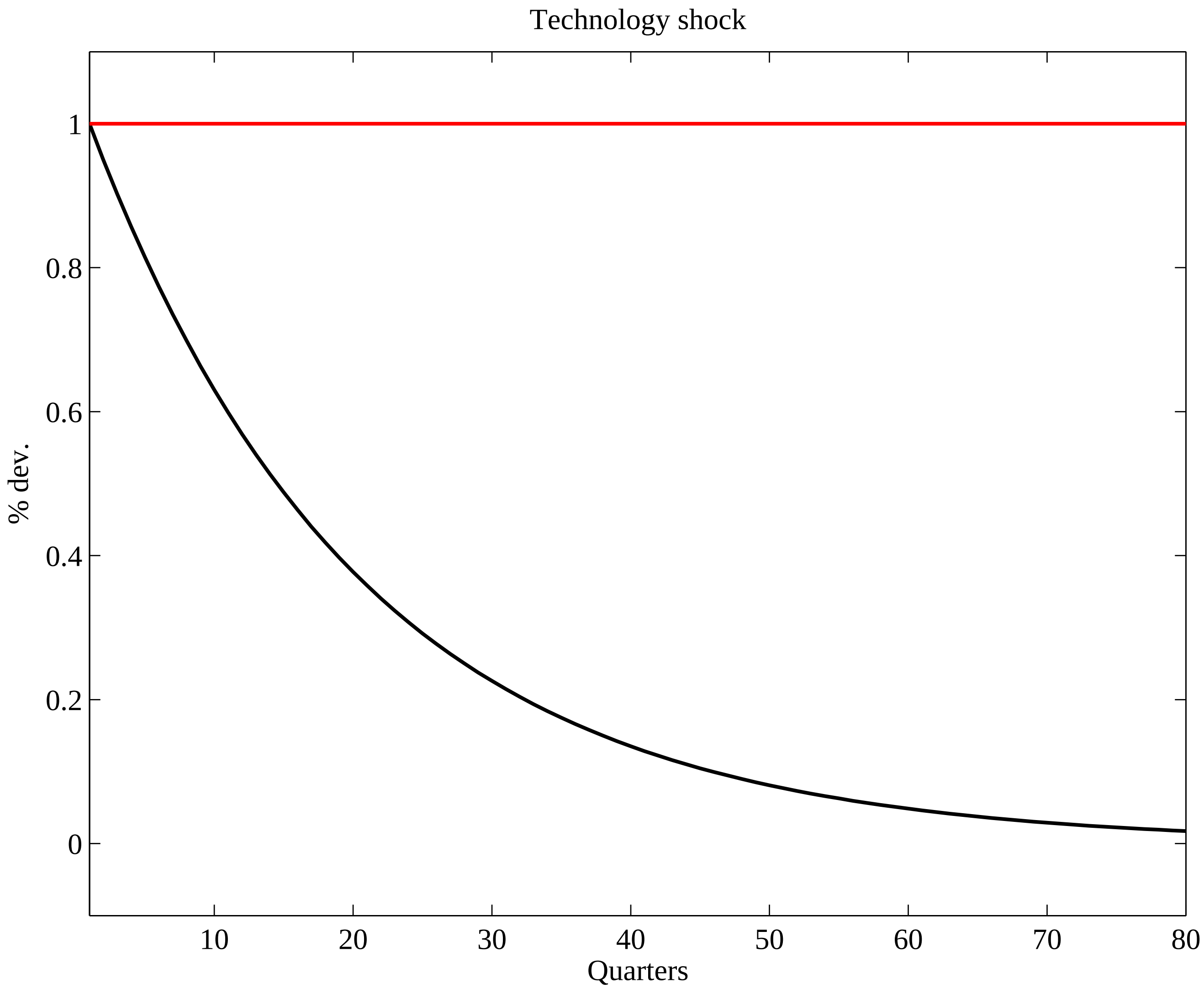
IRF to A Technological Shock: Temporary *vs* Persistent



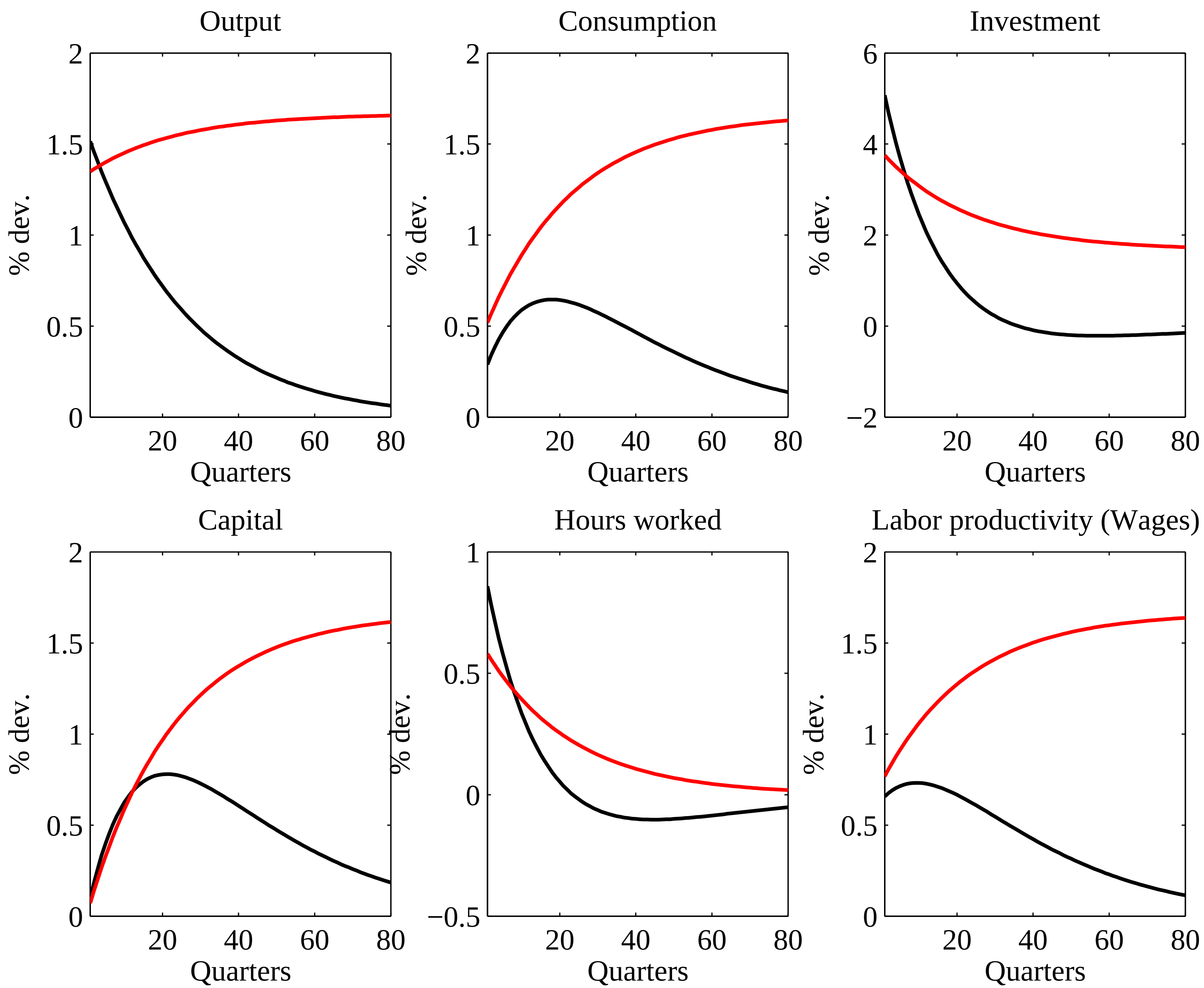
IRF to A Technological Shock: Temporary *vs* Persistent



IRF to A Technological Shock: Persistent *vs* Permanent



IRF to A Technological Shock: Persistent *vs* Permanent



- How to improve the model:
  - × Introducing additional shocks
  - × Improving the propagation mechanisms

3.9 Solving the puzzles

3.9.1 Adding Demand Shocks

- Government Expenditures

- Change the Household's budget constraint:

$$B_{t+1} + C_t + I_t + T_t \leqslant (1 + r_{t-1})B_t + W_t h_t + z_t K_t$$

- Government Balanced Budget: $T_t = G_t$

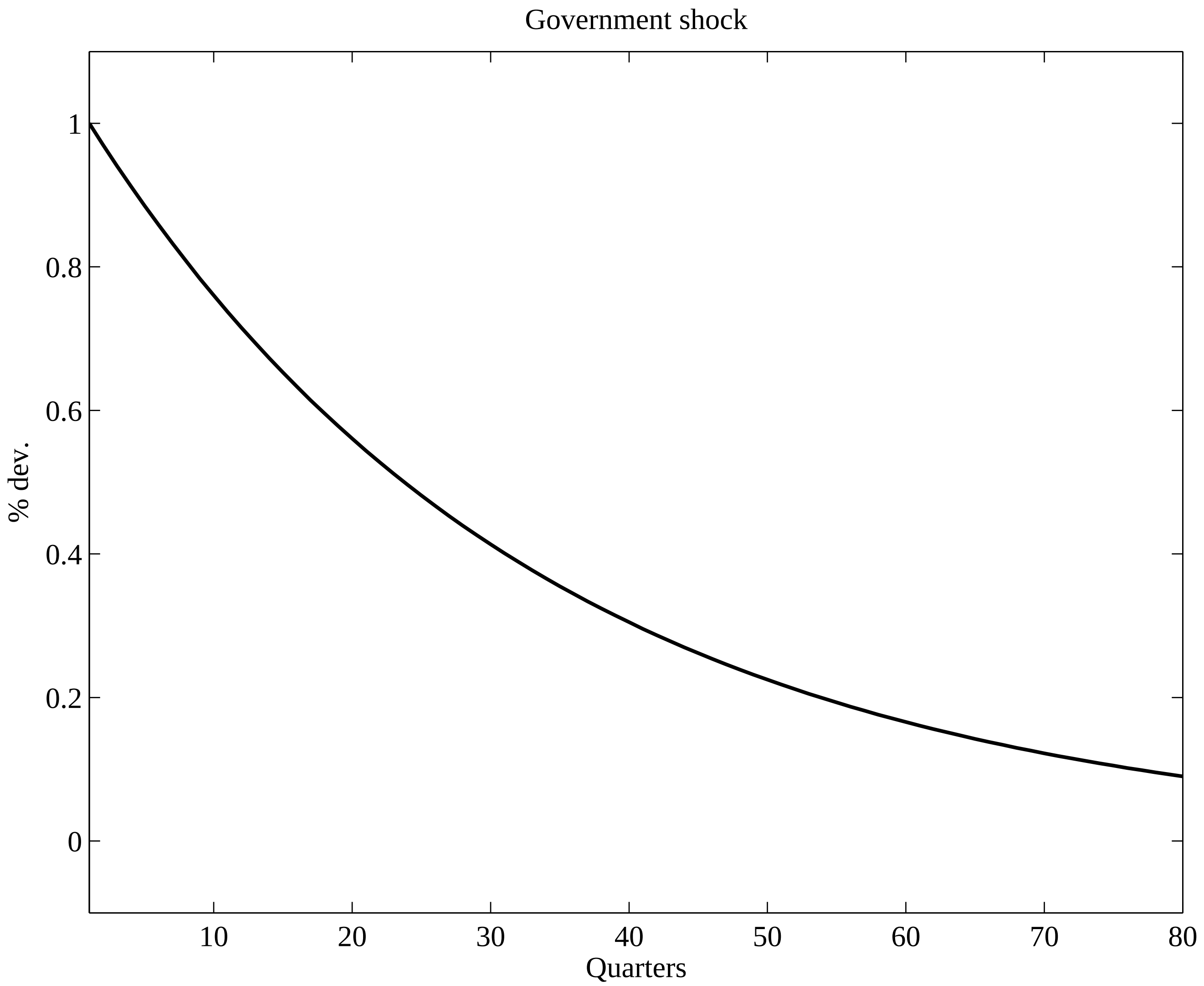
- Government expenditures are modeled as

$$\log(G_t) = \rho \log(G_{t-1}) + (1 - \rho_g) \log(\overline{G}) + \varepsilon_{g,t}$$

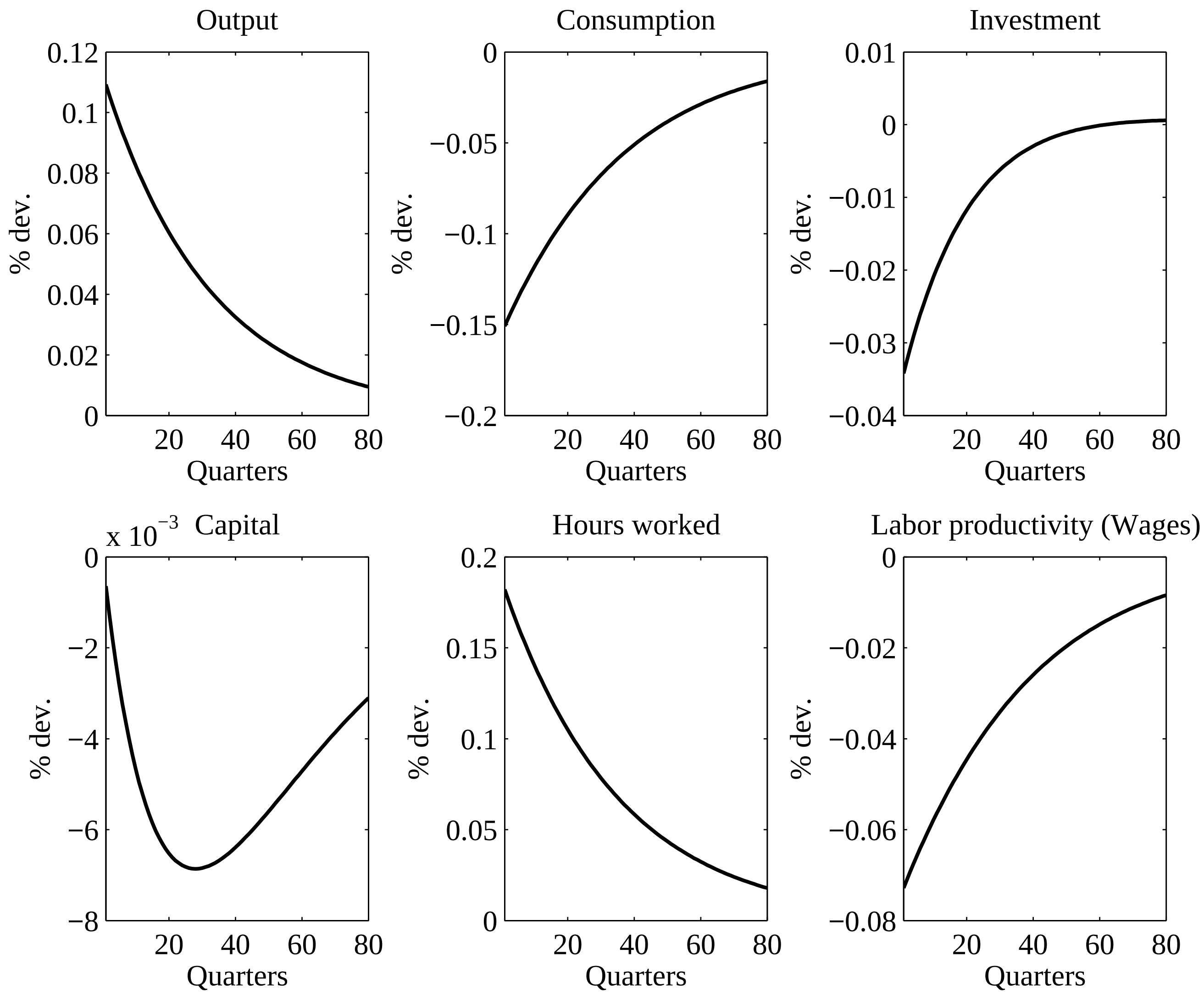
with $\varepsilon_{g,t} \rightsquigarrow \mathscr{N}(0, \sigma_t^2)$.

- $G/Y = 0.2$, $\rho_g = 0.97$, $\sigma_g = 0.02$.

IRF to A Government Spending Shock



IRF to A Government Spending Shock



Technological and Government Spending Model

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot,y)$	$\rho(\cdot,h)$	
Output	1.70	–	–	–	0.84
	1.43	–	–	–	0.68
Consumption	0.80	0.47	0.78	–	0.83
	0.62	0.43	0.54	–	0.75
Investment	6.49	3.83	0.84	–	0.81
	4.62	3.24	0.97	–	0.68
Hours worked	1.69	1.00	0.86	–	0.89
	0.85	0.59	0.90	–	0.68
Labor productivity	0.90	0.53	0.41	0.09	0.69
	0.76	0.53	0.87	0.58	0.72

(model in yellow)

3.9.2 Labor indivisibility

- Rogerson [1984] and Hansen [1985]: work a fixed amount of hours or does not (indivisibility)
- Then, the commodity set is nonconvex. We convexify it by the use of lotteries
- The nice property of this example is that one can define a representative agent, whose preferences are different from the original agents

- Preferences

$$U(C_{it}, 1-h_{it}) = \begin{cases} \log(C^e_{it}) + \theta \log(1-h_0) & \text{if she works} \\ \log(C^u_{it}) + \theta \log(1) & \text{if not} \end{cases}$$

- Randomly drawn to work with probability $\pi_t = \frac{h_t}{h_0}$

$$\begin{aligned} EU_t &= \pi_t \left(\log(C^e_{it}) + \theta \log(1-h_0)\right) + (1-\pi_t)\left(\log(C^u_{it}) + \theta \log(1)\right) \\ &= \pi_t \log(C^e_{it}) + (1-\pi_t)\log(C^u_{it}) + \theta \pi_t \log(1-h_0) \\ &= \pi_t \log(C^e_{it}) + (1-\pi_t)\log(C^u_{it}) + \theta \frac{\log(1-h_0)}{h_0} h_t \end{aligned}$$

- Households have access to a insurance contract that pays one unit of good in unemployed, 0 if employed.

- They buy A_{it} units of insurance, at unit price τ_t

$$\begin{cases} C_{it}^e + \tau_t A_{it} + K_{it+1}^e \leqslant (z_t + 1 - \delta) K_{it} + w_t h_0 & \text{if she works} \\ C_{it}^u + \tau_t A_{it} + K_{it+1}^u \leqslant (z_t + 1 - \delta) K_{it} + A_{it} & \text{if not} \end{cases}$$

- Households cannot manipulate the probability of working

- Risk neutral insurance companies will make zero profit: Revenues $\tau_t A_t$ = Costs $(1 - \pi_t) A_t$

- Therefore $\tau_t = 1 - \pi_t$

- Households will demand insurance so as to equalize marginal utility of consumption in the two states (employed and unemployed)

- Therefore $C_t^e = C_t^u$ (because utility is additively separable in consumption and leisure)

- We have (assuming they make same accumulation choice):

$$\begin{cases} C_t^e = -K_{t+1} + (z_t + 1 - \delta)K_{it} + w_t h_0 - (1-\pi_t)A_t & \text{if she works} \\ C_t^u = -K_{t+1} + (z_t + 1 - \delta)K_{it} + \pi_t A_{it} & \text{if not} \end{cases}$$

- Equalization of C^u and C^e gives $A = wh_0$.

- As consumption is the same and utility separable, accumulation decisions will be similar

- Assume that agents had initially the same wealth (in period 0): they will always take similar consumption and saving decisions

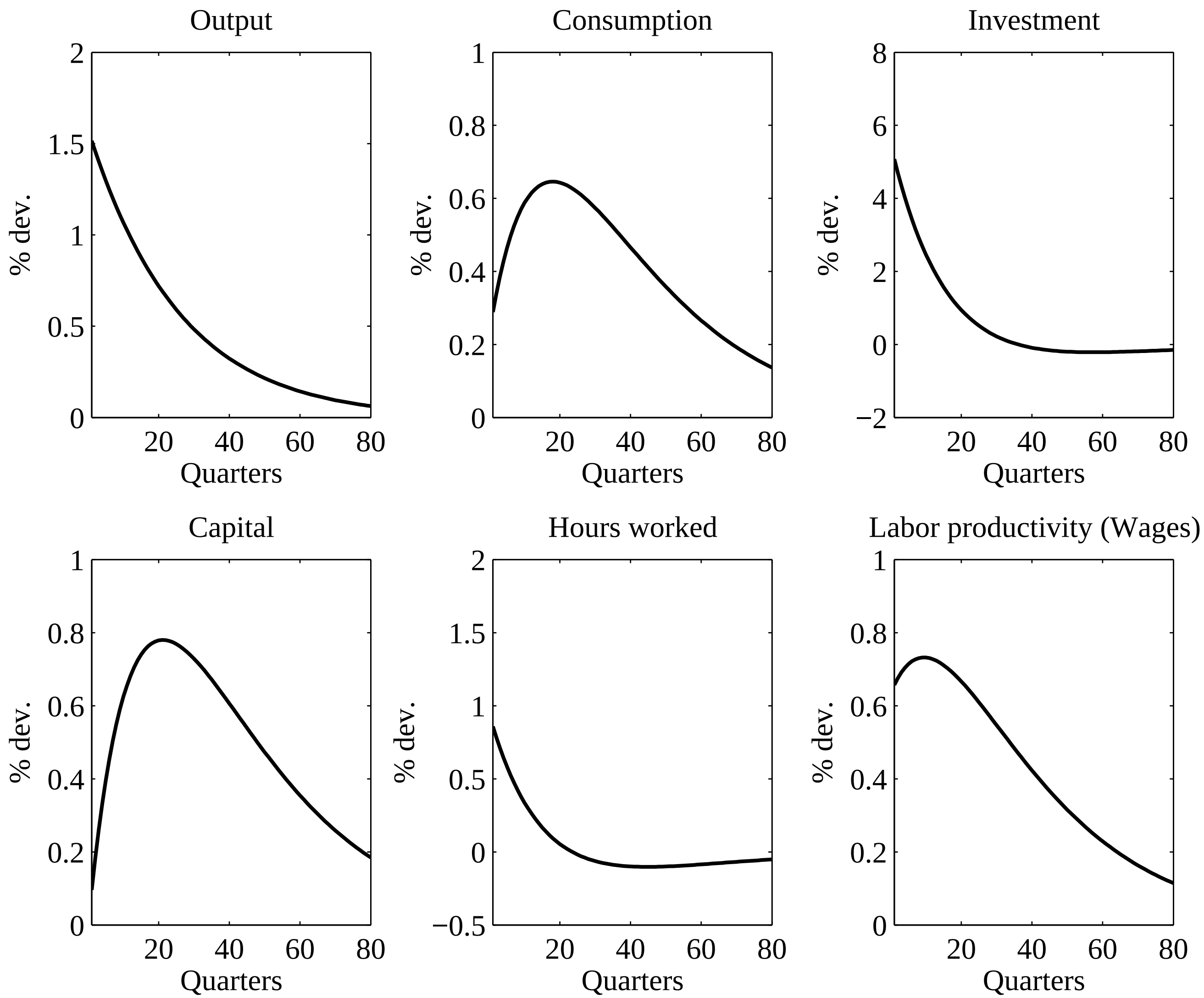
- An agent utility in t can therefore be written $\pi_t(\log C_t + \theta \log(1-h_0)) + (1-\pi_t)(\log C_t + \theta \log 1)$
- Utility collapses to

$$U(C_t, 1-h_t) = \log(C_t) - Bh_t$$

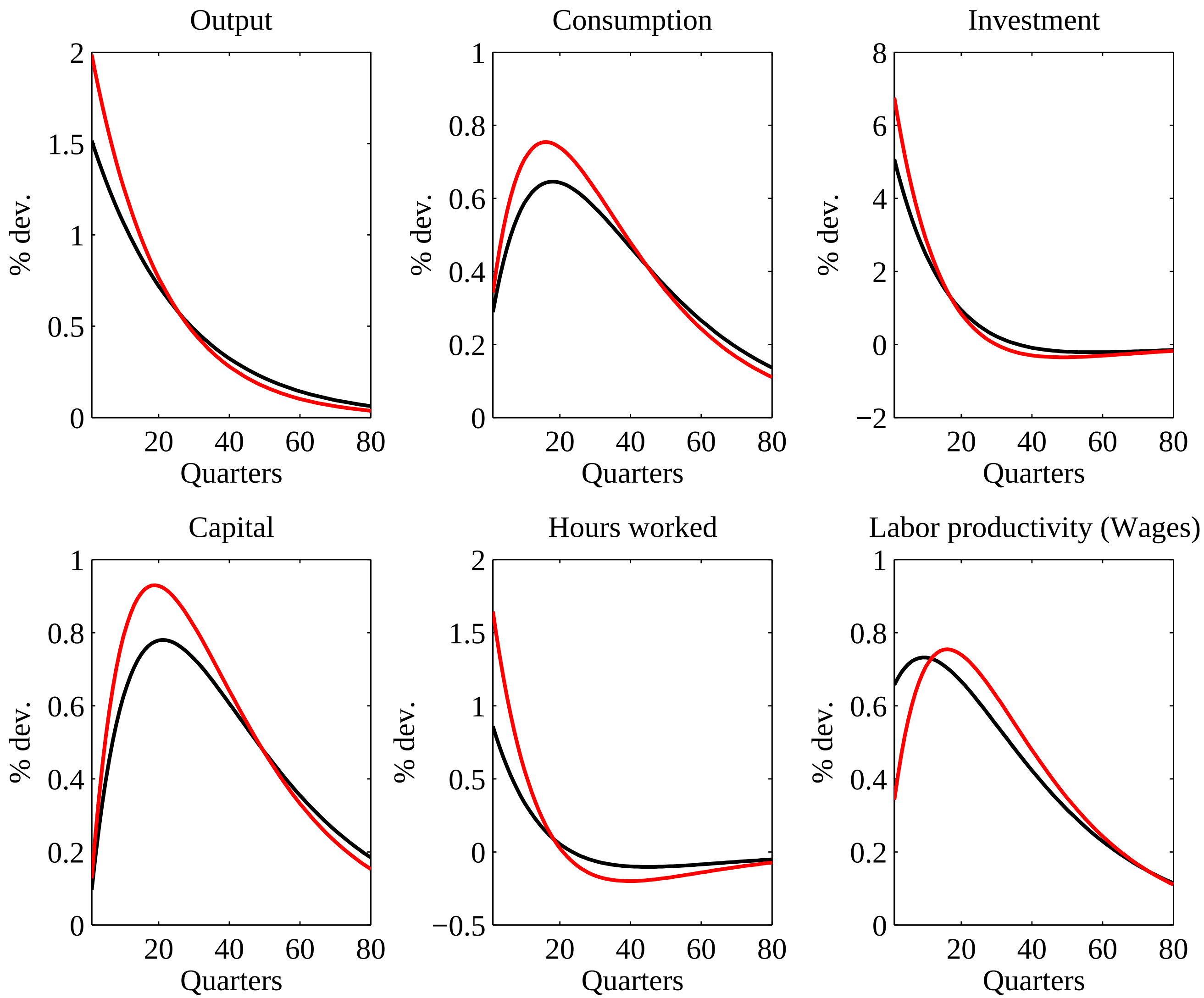
with $B = \theta \frac{\log(1-h_0)}{h_0}$

- The rest of the model (firms, equilibrium equations) remains unchanged

IRF to A Technological Shock - Indivisible Labor Model



IRF to A Technological Shock - Indivisible Labor Model



Indivisible Labor Model

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	
Output	1.70	–	–	–	0.84
	1.95	–	–	–	0.68
Consumption	0.80	0.47	0.78	–	0.83
	0.45	0.23	0.78	–	0.83
Investment	6.49	3.83	0.84	–	0.81
	6.66	3.40	0.99	–	0.67
Hours worked	1.69	1.00	0.86	–	0.89
	1.63	0.83	0.98	–	0.67
Labor productivity	0.90	0.53	0.41	0.09	0.69
	0.45	0.23	0.77	0.65	0.83

(model in yellow)

4 Business Cycle Accounting

Chari, Kehoe and McGrattan, "Business Cycle Accounting", Econometrica, Vol. 75, No. 3, 2007

4.1 Benchmark Economy

4.1.1 Agents Problems

- Notation: s_t are events, s^t are histories, $\pi_t(s^t)$ is probability of s^t as of period 0.
- Four wedges, that are four exogenous stochastic variables functions of the underlying random variable s^t

1. Efficiency wedge $A_t(s^t)$
2. Labor wedge $1 - \tau_{lt}(s^t)$
3. Investment wedge $1/[1 + \tau_{xt}(s^t)]$
4. Government consumption wedge $g_t(s^t)$

- Consumers maximize

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t) U(c_t(s^t), l_t(s^t)) N_t$$

subject to

$$c_t(s^t) + [1 + \tau_{xt}(s^t)] x_t(s^t) = [1 - \tau_{lt}(s^t)] w_t(s^t) l_t(s^t) + r_t(s^t) k_t(s^{t-1})$$

$$+T_t(s^t)$$

$$(1+\gamma_n)k_{t+1}(s^t) = (1-\delta)k_t(s^{t-1}) + x_t(s^t)$$

- Firms maximize

$$A_t(s^t)F(k_t(s^{t-1}),(1+\gamma)^t l_t(s^t)) - r_t(s^t)k_t(s^{t-1}) - w_t(s^t)l_t(s^t).$$

4.1.2 Equilibrium

- Equilibrium is summarized by

$$
\begin{aligned}
y_t(s^t) &= c_t(s^t) + x_t(s^t) + g_t(s^t), && (10)\\
y_t(s^t) &= A_t(s^t)F(k_t(s^{t-1}), (1+\gamma)^t l_t(s^t)), && (11)\\
\frac{U_{lt}(s^t)}{U_{ct}(s^t)} &= [1-\tau_{lt}(s^t)]A_t(s^t)(1+\gamma)^t F_{lt}, && (12)
\end{aligned}
$$

$$U_{ct}(s^t)[1+\tau_{xt}(s^t)] =$$

$$\beta \sum_{s^{t+1}} \pi_t(s^{t+1}|s^t) U_{ct+1}(s^{t+1}) \left\{ A_{t+1}(s^{t+1}) F_{kt+1}(s^{t+1}) \right. \tag{13}$$

$$\left. +(1-\delta)[1+\tau_{xt+1}(s^{t+1})] \right\}$$

- Note that one wedge (at least) enters in each of the four equations.

Remark : Those wedges are reduced forms for many imperfections:

Efficiency Wedge : Technological regression (?), mismeasurement, disorganization

Labor Wedge : Taxes, Unions, wages rigidities, implicit contracts, ...

Investment Wedge : Credit market imperfections,

Government Wedge : Government spending shocks, net exports shocks

4.2 Accounting procedure

- Consider (y_t, c_t, l_t, x_t).

- One can always find a sequence of the four wedges such that the model outcome exactly matches observed (y_t, c_t, l_t, x_t)
- The procedure is implemented on the US Great Depression
- As the government wedge plays no role for y, x and l, it is set to zero and we only consider fluctuations in y, x and l.

4.3 Results

- The 3 wedges are measured so that one exactly reproduces the movements of Y, I and L

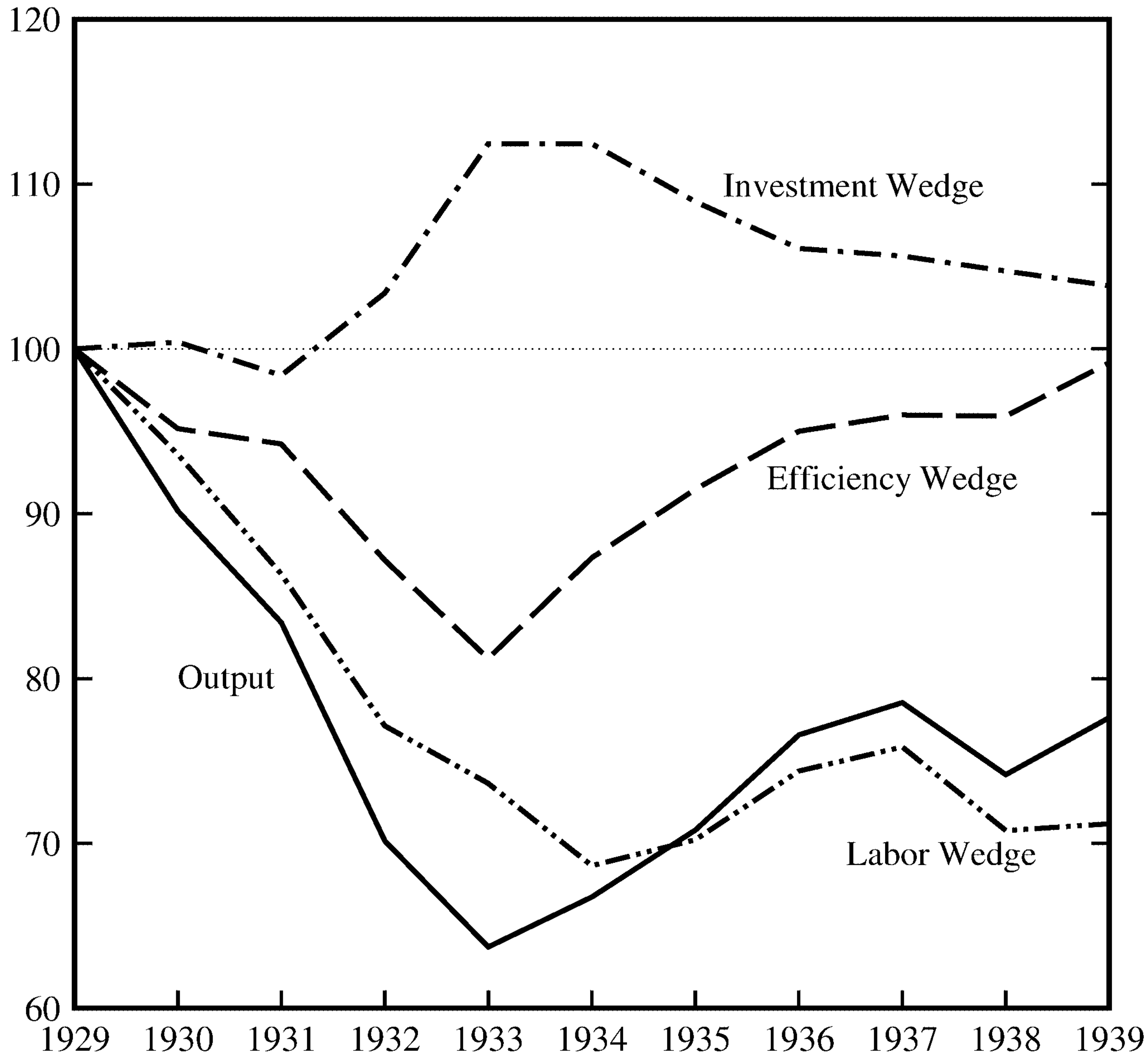


FIGURE 1.—U.S. output and three measured wedges (annually; normalized to equal 100 in 1929).

1. The efficiency wedge is in 1933 17 % below the 1929 level, but is back to normal level by 1939.

2. The labor wedge remains 30% below (permanently ?)

3. The investment wedge goes the wrong way

- Let us look at the fraction of macroeconomic movements that are explained by those 2 wedges, then by only the investment one, then by all but one wedge:

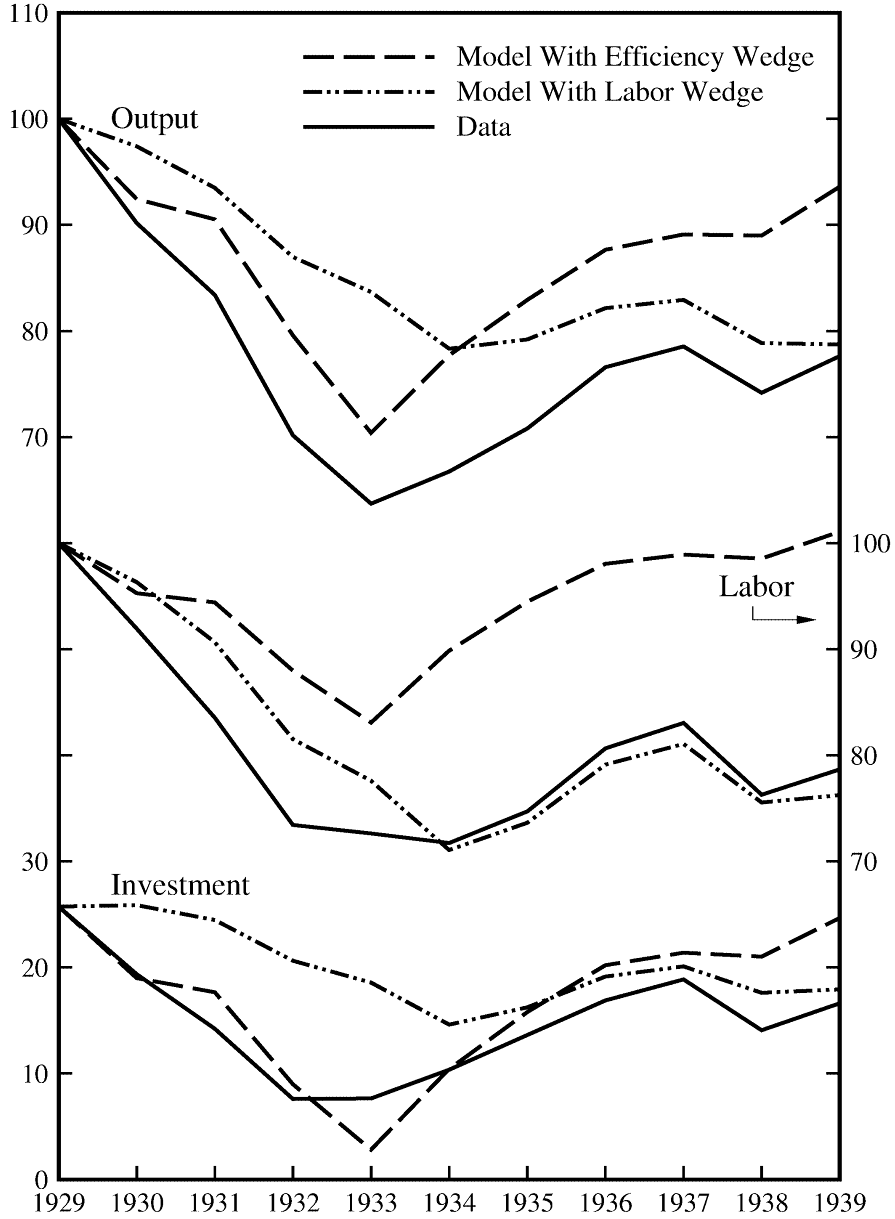


FIGURE 2.—Data and predictions of the models with just one wedge.

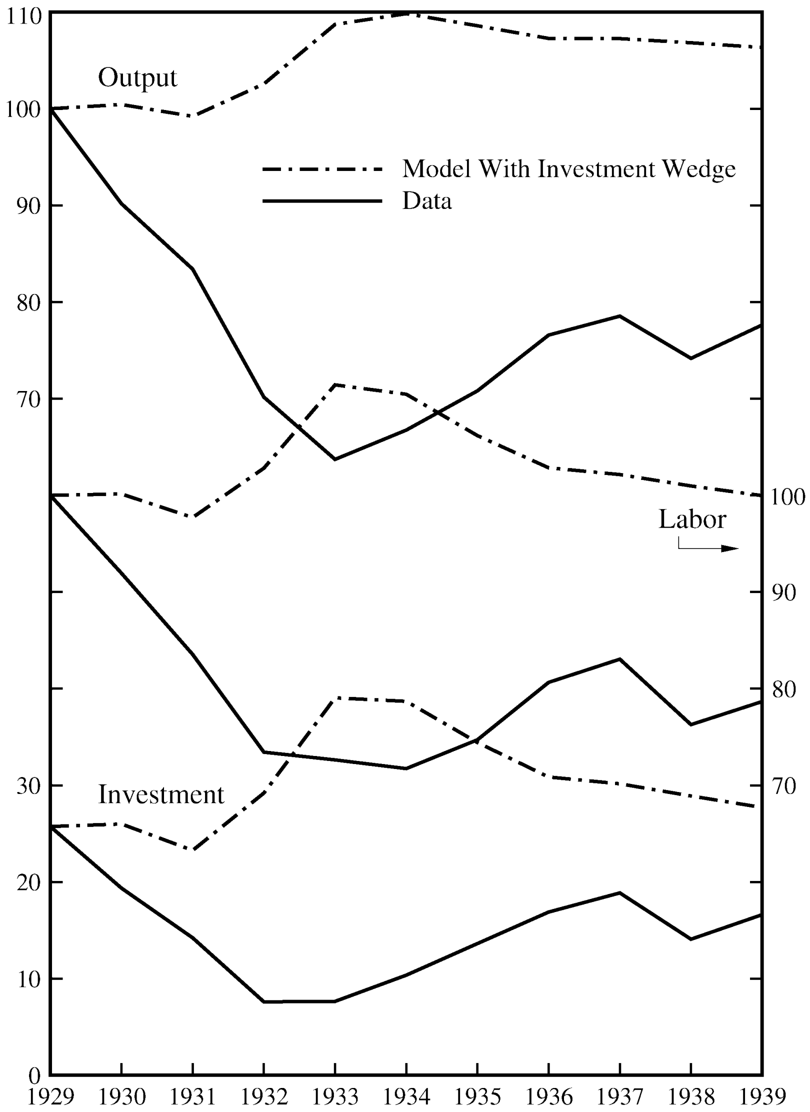


FIGURE 3.—Data and predictions of the model with just the investment wedge.

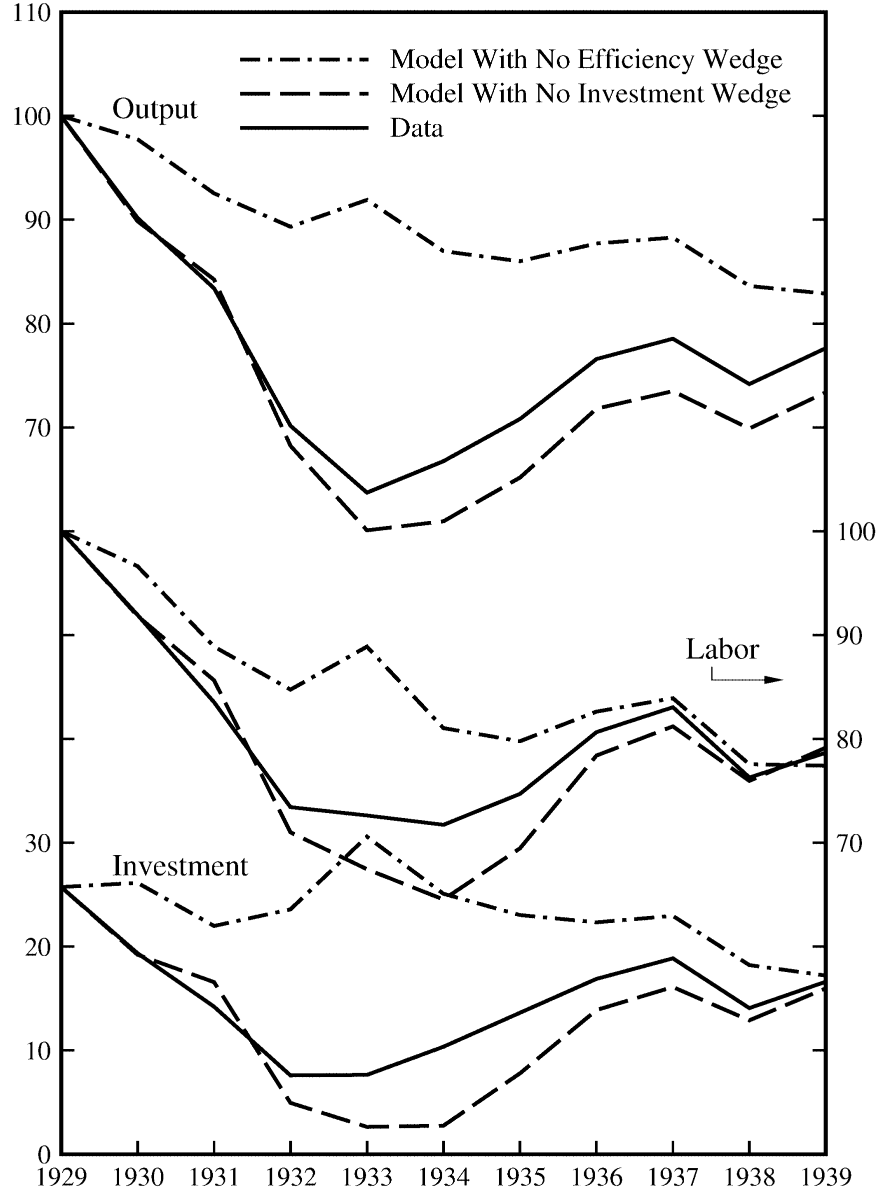


FIGURE 4.—Data and predictions of the models with all wedges but one.

1. With the efficiency wedge only: much of the downturn is accounted for, bit the recovery is too important

2. The efficiency wedge misses labor market outcome

3. With the labor market wedge only, the downturn is partially missed, but the absence of recovery is well accounted for.

4. labor market outcomes are well accounted by the labor wedge.

- With the 2 wedges, one gets almost all of the action (the investment wedge is not so important)

4.4 Cole and Ohanian Explanation

Cole and Ohanian, "New Deal Policies and the Persistence of the Great Depression: A General Equilibrium Analysis," Journal of Political Economy, 2004

- The recovery from the Great Depression was weak.
- Puzzling: the large negative shocks that some economists believe caused the 1929-33 downturn - including monetary shocks, productivity shocks, and banking shocks - become positive after 1933.
- Thesis: Roosevelt's "New Deal" cartelization policies, which limited competition in product markets and increased labor bar-

gaining power, kept the economy depressed after 1933.

- New Deal $\rightsquigarrow$ National Industrial Recovery Act (NIRA):

  1. suspended antitrust law and permitted collusion in some sectors...

  2. provided that industry raised wages above market clearing levels and accepted collective bargaining with independent labor unions.